Standard Form of a Quadratic Function

Name _____

Algebra 2 Lesson 2-2 (fast track) Date _____ Period _____

Learning goal: I can graph a quadratic function in standard form from its vertex, axis of symmetry and y-intercept, and use the vertex to find a maximum or minimum in a real-world problem.

Language goal: I can explain how I found a maximum or minimum using the words *standard form*, *vertex*, *axis of symmetry*, *y-intercept* and *maximum / minimum*.

On the quiz: (1) Find the minimum point of a parabola from its equation (Example 2, Step 1) (full Savvas quiz only). (2) Find the price change that gives a toy maker his highest profit, and that profit (Example 3; bonus). (3) Find the maximum height of a pebble tossed from a cliff (Example 3, Try It 3). (4) Pick the equation of a parabola through three points (full Savvas quiz only). (5) Graph a diver's path from the diving board to the water (Example 2, Try It 2, Practice).

On the Topic 2 test: Question 5 asks you to graph a rock dropped into water (Example 2). Question 6 asks you to graph a launch from a tower and give its maximum height (Examples 2 and 3). Question 3 (bottom of a glass) comes from Example 1 and Question 13 (parabola through the origin and two matching points) from Example 4, which this packet skips.

Example 2: Graph a Quadratic Function in Standard Form

How can you use key features to graph $f(x) = x^2 - 4x + 8$? **Standard form:** $y = ax^2 + bx + c$.

Identify a, b and c: $a = 1$, $b = -4$, $c = 8$.

Step 1 Vertex and axis of symmetry.

$$h = -\frac{b}{2a} = -\frac{(-4)}{2(1)} = 2$$

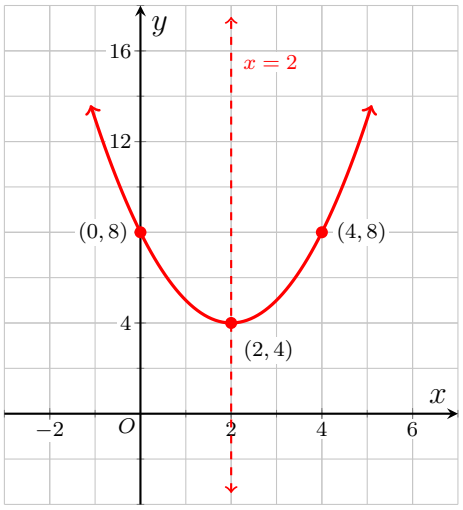
$$k = f(2) = (2)^2 - 4(2) + 8 = 4$$

Vertex $(2, 4)$; axis of symmetry $x = 2$.

Step 2 y**-intercept.** $f(0) = 0 - 0 + 8 = 8$, so $(0, 8)$. (It is always $(0, c)$.)

Step 3 Symmetric point. $(0, 8)$ is 2 units *left* of $x = 2$, so $(4, 8)$, 2 units *right*, is on the parabola too.

Step 4 Sketch. Plot the three points; $a = 1 > 0$, so the parabola opens up and the vertex is the **minimum**.



Say it: The **vertex** is (____, ____) and the **axis of symmetry** is $x =$ ____. The y**-intercept** $(0, 8)$ and the point (____, ____) are the same distance from the axis of symmetry, so . . .

Try It 2: Graph using key features

Use the key features to graph the function $f(x) = x^2 - 6x - 1$.

$a =$ _____ $b =$ _____ $c =$ _____

$h = -\dfrac{b}{2a} =$ _____

$k = f(h) =$ _____

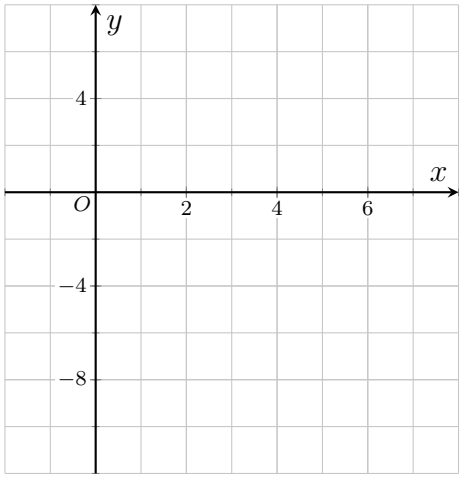
Vertex: (_____, _____)

Axis of symmetry: $x =$ _____

y-intercept: (_____, _____)

Symmetric point: (_____, _____)

Opens _____, so the vertex is a _____.



How did the axis of symmetry help you find a third point?

Example 3: Interpret the Graph of a Quadratic Function

The graph of the function $f(x) = -10x^2 + 700x - 6{,}000$ shows the profit a company earns for selling headphones at different prices. What is the maximum profit the company can expect to earn?

Formulate. The x-axis shows selling price and the y-axis shows the profit. The **maximum** y-value occurs at the **vertex** of the parabola.

Compute. $a = -10,\ \ b = 700.$

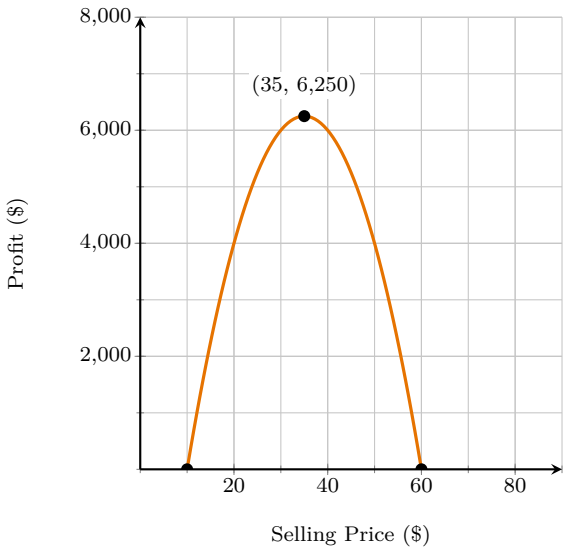
$$h = -\frac{b}{2a} = -\frac{700}{2(-10)} = -\frac{700}{-20} = 35$$

$$y = -10(35)^2 + 700(35) - 6{,}000$$
$$= -12{,}250 + 24{,}500 - 6{,}000 = 6{,}250$$

The vertex is $(35,\ 6{,}250)$.

Interpret. A selling price of \$35 per item gives the maximum profit of \$6,250.

Careful: there is a negative in the formula *and* a negative value of a.



Say it: $a = -10$ is negative, so the parabola opens _____ and the **vertex** is a _____. Its x-coordinate, _____, is the _____; its y-coordinate, _____, is the _____.

Try It 3: Maximum height

A water balloon is tossed straight up from the ground. The height of the water balloon over time can be modeled by the function $y = -14x^2 + 34x + 3$. What was the maximum height of the water balloon after it was tossed?

$a =$ _____ $b =$ _____ $c =$ _____

$h = -\dfrac{b}{2a} =$ _____

_____

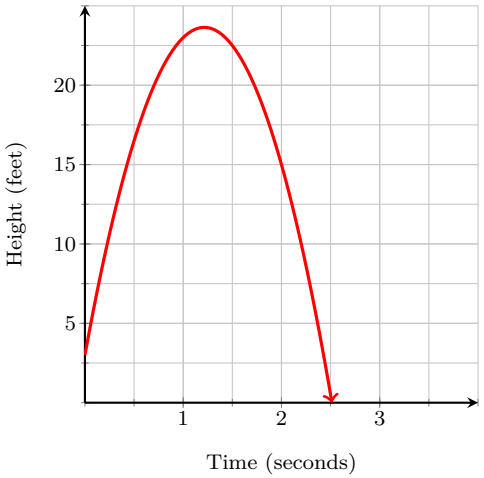
_____

$y =$ _____

_____

_____

Maximum height: about _____ ft, reached after about _____ s.



Is the vertex a maximum or a minimum? How do you know before you graph?

_____

_____

Practice: From the throw to the ground

A ball is thrown upward from a balcony. Its height, in meters, after x seconds is $y = -5x^2 + 10x + 15$. Graph the function from the point where the ball was thrown to the point where it hits the ground.

a. Starting point (y-intercept): (____, ____)

b. Vertex: $h =$ _____, $k =$ _____

c. Symmetric point: (____, ____)

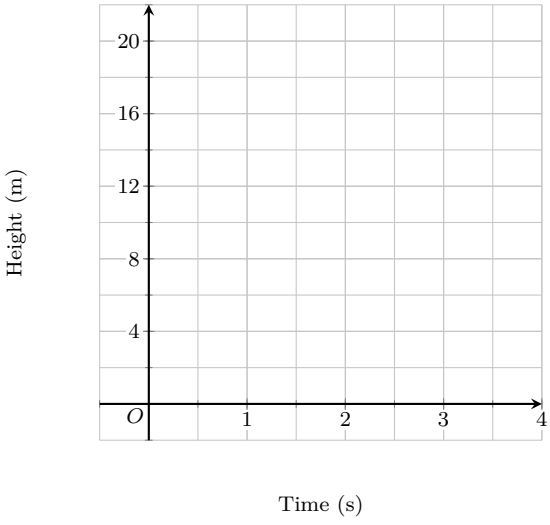
d. Keep going until the height is 0:

x	0.5	1.5	2.5	3
y				

e. Graph only from the throw to the ground: a closed dot at each end, no arrows.

f. How high is the balcony? What is the greatest height of the ball?

_____



IXL: Practice for the quiz

Go to IXL, Algebra 2, and type the code in the search box. Do the starred skills first.

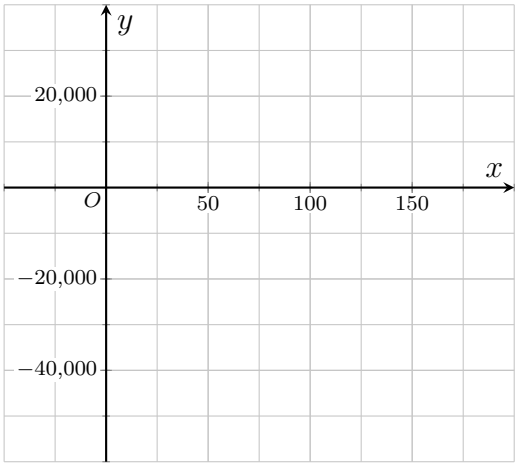
Code	Skill	Feeds
L8C ★	Characteristics of quadratic functions: equations	Quiz 3 and bonus Quiz 2 (Example 3)
S9G ★	Graph a quadratic function	Quiz 5: the diver's path (Example 2)
WMS	Characteristics of quadratic functions: graphs	Warm-up: key features off a graph
8GN	Multiply polynomials	Warm-up: expand $2(x-4)^2+5$

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Practice 26 (DOK 3)

A college's business office found the relationship between the number of admissions counselors they employ and the college's profit from tuition could be modeled by the function $y = -10x^2 + 1{,}500x - 35{,}000$.

a. Graph the function.
b. How many admissions counselors should the college employ to maximize its profit?



c. What is the maximum amount of profit the college can make?

Finish the sentences.
The **vertex** is (_____, _____). The parabola opens _____ because a is _____, so the vertex is a _____.
The college should employ _____ counselors, the x-coordinate of the vertex. The maximum profit is _____, the y-coordinate.